



Fourier series

Periodic functions occur quite frequently in engineering problems. Their representation in terms of simple periodic functions, such as, sine and cosine is a matter of great importance which leads to Fourier series. These series are a very powerful tool in solving various problems involving partial differential equations.

The theory of Fourier series is rather complicated, but the application of these series is simple. Fourier series are, in a certain sense, more universal than Taylor series, because many discontinuous periodic functions of practical interest can be developed in Fourier series, but, of-course, do not have Taylor series representation. We first give the following definitions which will be used in our later work.



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Periodic function

A function $f(x)$ is said to be periodic if there is some positive constant P , such that

$$f(x + P) = f(x), \quad \text{for all } x.$$

This constant P is then called a period of $f(x)$. In other words, if the value of each ordinate $f(x)$ repeats itself at equal intervals in the abscissa, then $f(x)$ is said to be a periodic function. The graph of a such function is obtained by periodic repetition of its graph in any interval of length P . A function which is periodic with period P is shown graphically in the following figure.

Figure

It is easy to verify that if P is a period of $f(x)$, then $2P$ is also a period of $f(x)$ since from and, in-general, if n is a positive integer, then nP is also a period of $f(x)$, i.e.,

$$f(x + nP) = f(x), \quad \text{for all } x.$$



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Hence if P is a period of $f(x)$, then $2P, 3P, 4P, \dots$ are also periods of $f(x)$. The least period of $f(x)$ is often called the **fundamental period** or simply the period of $f(x)$.

Example:- The function $\sin x$ has periods: $2\pi, 4\pi, 6\pi, \dots$. Since

$$\sin(x + 2\pi) = \sin(x + 4\pi) = \dots = \sin x.$$

However, 2π is the least period or simply the period of $\sin x$. Similarly, period of $\cos x$ is 2π . while the period of $\tan x$ is π , since $\cos(x + 2\pi) = \cos x$ and $\tan(x + \pi) = \frac{\tan x + \tan \pi}{1 - \tan x \tan \pi} = \tan x$.

The fundamental period of $\sin nx$ and $\cos nx$, where n is a positive integer, is $\frac{2\pi}{n}$, since

$$\sin n\left(x + \frac{2\pi}{n}\right) = \sin(nx + 2\pi) = \sin nx \cos 2\pi + \cos nx \sin 2\pi = \sin nx,$$

$$\cos n\left(x + \frac{2\pi}{n}\right) = \cos(nx + 2\pi) = \cos nx \cos 2\pi - \sin nx \sin 2\pi = \cos nx.$$



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A constant function $f(x) = \text{constant} = C$ will also be regarded as a periodic function in the sense of definition, since for any positive number P , we have

$$f(x + P) = f(x) = C, \quad \text{for all } x.$$

Thus any positive number is a period of $f(x)$. Since there is no smallest positive number, $f(x)$ has no fundamental period.

Examples of periodic functions which arise in engineering and physical situations are: the propagation of electric current in an alternating current circuit, mechanical vibrations, wave propagation in different types of media, the displacement of a piston in a piston engine, the motion of the Earth about the Sun (to very high accuracy) with a period of one year, the bending of a telephone cable which is suspended from regularly spaced poles is a function which is periodic w.r.t. the distance. In Biology, the beating of heart under normal conditions is also an example of a periodic function.

Examples of functions that are not periodic: $x, x^2, x^3, e^x, \cosh x$, etc.



Fourier series

Advantages of Fourier series

- 1 Discontinuous function can be represented by Fourier series. Although derivatives of the discontinuous functions do not exist. (This is not true for Taylor's series).
- 2 The Fourier series is useful in expanding the periodic functions since outside the closed interval, there exists a periodic extension of the function.
- 3 Expansion of an oscillating function by Fourier series gives all modes of oscillation (fundamental and all overtones) which is extremely useful in physics.
- 4 Fourier series of a discontinuous function is not uniformly convergent at all points.
- 5 Term by term integration of a convergent Fourier series is always valid, and it may be valid if the series is not convergent. However, term by term, differentiation of a Fourier series is not valid in most cases.



Fourier series

Orthogonality relationships

We now give some important integrals which will be used in the subsequent analysis.

Theorem:- Prove that

$$\left. \begin{array}{l} i) \quad \int_{-\pi}^{\pi} \sin kx \, dx = 0 \\ ii) \quad \int_{-\pi}^{\pi} \cos kx \, dx = 0 \end{array} \right\}, \quad \text{if } k = 1, 2, 3, \dots$$

Proof:- i)

$$\begin{aligned} \int_{-\pi}^{\pi} \sin kx \, dx &= -\frac{1}{k} \left[\cos kx \right]_{-\pi}^{\pi} \\ &= -\frac{1}{k} [\cos kx - \cos(-kx)] \\ &= -\frac{1}{k} [\cos kx - \cos(kx)] = 0. \end{aligned}$$



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Theorem:- Prove that

$$i) \quad \int_{-\pi}^{\pi} \cos mx \cos nx \, dx = \begin{cases} 0, & \text{if } m \neq n \\ \pi, & \text{if } m = n \end{cases}$$

$$ii) \quad \int_{-\pi}^{\pi} \sin mx \sin nx \, dx = \begin{cases} 0, & \text{if } m \neq n \\ \pi, & \text{if } m = n \end{cases}$$

$$iii) \quad \int_{-\pi}^{\pi} \sin mx \cos nx \, dx = 0, \text{ for all } m, n, \text{ where } m, n = 1, 2, 3, \dots$$



Fourier series

Property of a periodic function

We now prove a result which will be useful in our later work.

Theorem:- Let $f(x)$ be a periodic function with period 2π , then

$$\int_{-\pi}^{\pi} f(x) dx = \int_C^{C+2\pi} f(x) dx, \quad \text{where } C \text{ is an arbitrary constant.}$$

Proof:- We have

$$\int_C^{C+2\pi} f(x) dx = \int_C^{-\pi} f(x) dx + \int_{-\pi}^{\pi} f(x) dx + \int_{\pi}^{C+2\pi} f(x) dx \quad (33)$$

Let $x = y + 2\pi$ in the last integral on the right of equation (33), then

$$\begin{aligned} \int_{\pi}^{C+2\pi} f(x) dx &= \int_{-\pi}^C f(y + 2\pi) dy \\ &= - \int_C^{-\pi} f(y + 2\pi) dy \end{aligned}$$



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$$\begin{aligned}\int_{\pi}^{C+2\pi} f(x) dx &= - \int_C^{-\pi} f(x+2\pi) dx \\ &= - \int_C^{-\pi} f(x) dx \quad \because f(x+2\pi) = f(x)\end{aligned}$$

Thus from equation (33), we have

$$\begin{aligned}\int_C^{C+2\pi} f(x) dx &= \int_C^{-\pi} f(x) dx + \int_{-\pi}^{\pi} f(x) dx - \int_C^{-\pi} f(x) dx \\ \text{or } \int_{-\pi}^{\pi} f(x) dx &= \int_C^{C+2\pi} f(x) dx.\end{aligned}$$

This theorem states that if $f(x)$ is a periodic function with period 2π , the interval of integration may be replaced by any other interval of length 2π .



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Note:- If $C = 0$, then we have

$$\int_{-\pi}^{\pi} f(x) dx = \int_0^{2\pi} f(x) dx.$$

Definition of Fourier series

Let $f(x)$ be a periodic function of period 2π defined in the interval $[-\pi, \pi]$. The infinite geometric series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

is called the **Fourier series** or **Fourier expression** of $f(x)$ if the coefficients a_n and b_n are given by



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$$\left. \begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \end{aligned} \right\}, \quad n = 1, 2, 3, \dots \quad (34)$$

The function $f(x)$ is called the **defining** or **generating** function of the Fourier series and the coefficients a_n and b_n given by equation (34) are known as Euler-Fourie coefficients.



Fourier series

Change of interval

Since $f(x)$ is periodic function with period 2π , $f(x) \cos nx$ and $f(x) \sin nx$ are also periodic. Hence, the Fourier coefficients a_n and b_n can be determined equivalently from

$$\left. \begin{aligned} a_0 &= \frac{1}{\pi} \int_C^{C+2\pi} f(x) dx \\ a_n &= \frac{1}{\pi} \int_C^{C+2\pi} f(x) \cos nx dx \\ b_n &= \frac{1}{\pi} \int_C^{C+2\pi} f(x) \sin nx dx \end{aligned} \right\}, \quad n = 1, 2, 3, \dots, \quad (35)$$

where C is some arbitrary constant. In the special case when $C = -\pi$, equation (35) becomes equation (34).



Fourier series

If $C = 0$ in equation (35), then we have

$$\left. \begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx \\ a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \\ b_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \end{aligned} \right\}, \quad n = 1, 2, 3, \dots \quad (36)$$

The coefficients in equation (36) are useful in computing Fourier series if the generating function $f(x)$ is defined in the interval $[0, 2\pi]$.



Fourier series

Example:-

Find the Fourier series of the following functions which are periodic with period 2π .

$$i) \quad f(x) = \begin{cases} 0 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

$$ii) \quad f(x) = \begin{cases} 1 & \text{if } 0 < x < \pi \\ 0 & \text{if } \pi < x < 2\pi \end{cases}$$

$$iii) \quad f(x) = \begin{cases} 1 & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ -1 & \text{if } \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

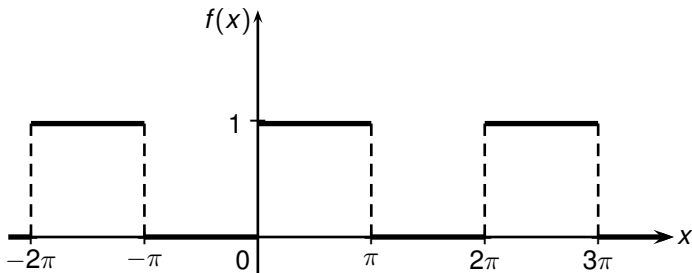


Fourier series

Solution:- i)

$$f(x) = \begin{cases} 0 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

The graph of $f(x)$ with period 2π and its periodic extension is shown in the following figure.





Fourier series

The Fourier series corresponding to $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (37)$$

where

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) dx + \int_0^{\pi} f(x) dx \right] \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} 1 dx \right] = \frac{1}{\pi} \int_0^{\pi} 1 dx = \frac{1}{\pi} |x|_0^{\pi} = 1 \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) \cos nx dx + \int_0^{\pi} f(x) \cos nx dx \right] \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 (0) \cos nx dx + \int_0^{\pi} (1) \cos nx dx \right] \end{aligned}$$



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$$\begin{aligned}a_n &= \frac{1}{\pi} \int_0^{\pi} \cos nx \, dx = \frac{1}{n\pi} \left| \sin nx \right|_0^{\pi} \\&= \frac{1}{n\pi} (\sin n\pi) = 0 \quad (\text{since } \sin n\pi = 0 \text{ always})\end{aligned}$$

$$\begin{aligned}b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) \sin nx \, dx + \int_0^{\pi} f(x) \sin nx \, dx \right] \\&= \frac{1}{\pi} \left[\int_{-\pi}^0 (0) \sin nx \, dx + \int_0^{\pi} (1) \sin nx \, dx \right] \\&= \frac{1}{\pi} \int_0^{\pi} \sin nx \, dx = -\frac{1}{n\pi} \left| \cos nx \right|_0^{\pi} \\&= \frac{1}{n\pi} (1 - \cos n\pi) = \frac{1}{n\pi} (1 - (-1)^n)\end{aligned}$$

Thus $b_1 = \frac{2}{\pi}$, $b_2 = 0$, $b_3 = \frac{2}{3\pi}$, $b_4 = 0$, $b_5 = \frac{2}{5\pi}, \dots$



Fourier series

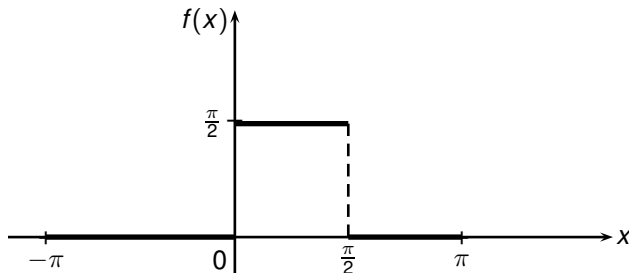
Hence from equation (37), the required Fourier series is given by

$$\begin{aligned}f(x) &= \frac{1}{2} + \left(\frac{2}{\pi} \sin x + \frac{2}{3\pi} \sin 3x + \frac{2}{5\pi} \sin 5x + \cdots \right) \\&= \frac{1}{2} + \frac{2}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \cdots \right)\end{aligned}$$



Fourier series

Example:- Find the Fourier series of the function shown in the following figure which is periodic with period 2π .





Fourier series

Solution:- From the figure, we have

$$f(x) = \begin{cases} 0 & \text{if } -\pi < x < 0 \\ \frac{\pi}{2} & \text{if } 0 < x < \frac{\pi}{2} \\ 0 & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$$

Since the period is 2π , therefore the Fourier series corresponding to $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



Fourier series

Functions having arbitrary period $P = 2L$

The functions discussed so far had period 2π , whereas most functions in applications will have periods other than 2π . We therefore obtain a Fourier series for a function $f(x)$ which is defined over the interval $[-L, L]$ and has the period $P = 2L$, where L is some arbitrary number.

Theorem:-

Prove that the Fourier series of a function $f(x)$ which is periodic with period $P = 2L$ and defined over the interval $[-L, L]$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{L}x + b_n \sin \frac{n\pi}{L}x \right)$$

where
$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$



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$$\left. \begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi}{L} x \, dx \\ b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi}{L} x \, dx \end{aligned} \right\}, \quad n = 1, 2, 3, \dots \quad (38)$$

Change of interval

Since $f(x)$ is periodic function with period $2L$, $f(x) \cos \frac{n\pi}{L} x$ and $f(x) \sin \frac{n\pi}{L} x$ are also periodic with period $2L$. Hence, the Fourier coefficients a_n and b_n can be determined equivalently from

$$a_0 = \frac{1}{L} \int_C^{C+2L} f(x) \, dx$$



Fourier series

$$\left. \begin{aligned} a_n &= \frac{1}{L} \int_C^{C+2L} f(x) \cos \frac{n\pi}{L} x \, dx \\ b_n &= \frac{1}{L} \int_C^{C+2L} f(x) \sin \frac{n\pi}{L} x \, dx \end{aligned} \right\}, \quad n = 1, 2, 3, \dots, \quad (39)$$

where C is some arbitrary constant. In the special case when $C = -L$, equation (39) becomes equation (38). If $C = 0$ in equation (39), then we have

$$\left. \begin{aligned} a_0 &= \frac{1}{L} \int_0^{2L} f(x) \, dx \\ a_n &= \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi}{L} x \, dx \\ b_n &= \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi}{L} x \, dx \end{aligned} \right\}, \quad n = 1, 2, 3, \dots. \quad (40)$$

The coefficients in equation (40) are useful in computing Fourier series if the generating function $f(x)$ is defined in the interval $[0, 2L]$.



Fourier series

Example:- Find the Fourier series of the following functions:

$$i) \quad f(x) = \begin{cases} -1 & \text{if } -1 < x < 0 \\ 1 & \text{if } 0 < x < 1 \end{cases} \quad P = 2L = 2$$

$$ii) \quad f(x) = \begin{cases} 0 & \text{if } -1 < x < 1 \\ 1 & \text{if } 1 < x < 3 \end{cases} \quad P = 2L = 4$$



Fourier series

Fourier series for even and odd functions

Unnecessary calculations in determining the Fourier coefficients can be avoided if the function is even or odd.

Even function

A function $f(x)$ is said to be even if

$$f(-x) = f(x), \quad \text{for all } x.$$

For example: x^2 , x^4 , $\cos nx$, and $x^6 - 2x^2 + \cos x$ are even functions.

Odd function

A function $f(x)$ is said to be odd if

$$f(-x) = -f(x), \quad \text{for all } x.$$

For example: x , x^3 , $\sin nx$, and $x^5 - 3x^3 + 2x$ are odd functions.



Fourier series

Remarks:-

The graph of an even function is symmetric about the y -axis while the graph of an odd function is symmetric about the origin.

A given function $f(x)$ not necessarily an even or odd. For example: e^x , $x^2 - x$, $\sin x + \cos x$, and $|x| + x$ are neither even nor odd. However, a given function $f(x)$ can be expressed as the sum of an even and an odd function since

$$\begin{aligned} f(x) &= \left[\frac{f(x) + f(-x)}{2} \right] + \left[\frac{f(x) - f(-x)}{2} \right] \\ &= \text{even function} + \text{odd function} \end{aligned}$$

$$\begin{aligned} \text{For example, } e^x &= \left(\frac{e^x + e^{-x}}{2} \right) + \left(\frac{e^x - e^{-x}}{2} \right) \\ &= \cosh x + \sinh x \\ &= \text{even function} + \text{odd function} \end{aligned}$$



Fourier series

Theorem:- Prove that

- i) the product of two even or two odd functions is even.
- ii) the product of an even and an odd function is odd.

Proof:- Let $h(x) = f(x)g(x)$

- i) If $f(x)$ and $g(x)$ are both even functions, then

$$h(-x) = f(-x)g(-x) = f(x)g(x) = h(x)$$

showing that their product is even.

- If $f(x)$ and $g(x)$ are both odd functions, then

$$h(-x) = f(-x)g(-x) = [-f(x)][-g(x)] = f(x)g(x) = h(x)$$

showing again that their product is even.



Fourier series

Example:-

Since x^2 and $\cos nx$ are both even functions while x^3 and $\sin nx$ are both odd functions, the product $x^2 \cos nx$ is even function. Similarly, $x^3 \sin nx$ is an even function while $x^2 \sin nx$ is an odd function.

Theorem:-

Let $f(x)$ be defined on the interval $[-a, a]$. Show that

i) If $f(x)$ is an even function, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$

ii) If $f(x)$ is an odd function, then

$$\int_{-a}^a f(x) dx = 0.$$



Fourier series

Proof:- Let

$$\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx. \quad (41)$$

Put $x = -t$ in the first integral on the right side of equation (41), then we have

$$\begin{aligned} \int_{-a}^0 f(x) dx &= - \int_a^0 f(-t) dt = \int_0^a f(-t) dt \\ &= \int_0^a f(-x) dx. \end{aligned}$$

Then from equation (41), we get

$$\int_{-a}^a f(x) dx = \int_0^a f(-x) dx + \int_0^a f(x) dx. \quad (42)$$



Fourier series

i) If $f(x)$ is an even function, then $f(-x) = f(x)$, and from equation (42), we get

$$\int_{-a}^a f(x) dx = \int_0^a f(x) dx + \int_0^a f(x) dx = 2 \int_0^a f(x) dx.$$

ii) If $f(x)$ is an odd function, then $f(-x) = -f(x)$, and from equation (42), we get

$$\int_{-a}^a f(x) dx = - \int_0^a f(x) dx + \int_0^a f(x) dx = 0.$$



Fourier series

Fourier series for an even function

Theorem:- If $f(x)$ is an even function which is also periodic with period $P = 2L$ and defined over the interval $[-L, L]$, then show that

$$a_0 = \frac{2}{L} \int_0^L f(x) dx$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi}{L} x dx, \quad b_n = 0, \quad n = 1, 2, 3, \dots,$$

and the corresponding Fourier series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{L} x.$$



Fourier series

Fourier series for an odd function

Theorem:- If $f(x)$ is an odd function which is also periodic with period $P = 2L$ and defined over the interval $[-L, L]$, then show that

$$a_0 = a_n = 0,$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x \, dx, \quad n = 1, 2, 3, \dots,$$

and the corresponding Fourier series is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{L} x.$$



Fourier series

Particular case of period 2π

i) If $f(x)$ is an even function which is also periodic with period $P = 2\pi$ and defined over the interval $[-\pi, \pi]$, then the Fourier series corresponding to $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx,$$

where $a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$

and $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx, \quad n = 1, 2, 3, \dots$



Fourier series

ii) If $f(x)$ is an odd function which is also periodic with period $P = 2\pi$ and defined over the interval $[-\pi, \pi]$, then the Fourier series corresponding to $f(x)$ is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx,$$

$$\text{where } b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx, \quad n = 1, 2, 3, \dots$$

Example:- Find the Fourier series of the following functions:

$$i) \quad f(x) = \begin{cases} -x & \text{if } -\pi < x < 0 \\ x & \text{if } 0 < x < \pi \end{cases}$$

$$ii) \quad f(x) = x, \quad (-\pi < x < \pi).$$



Fourier series

Half range Fourier series

In many problems of physics and engineering, there is a need of use the Fourier series of a function $f(x)$ that is defined only on the interval $[0, L]$. We know that the Fourier series of an even function is a Fourier cosine series containing only the cosine terms, whereas the Fourier series of an odd function is a Fourier sine series containing only the sine terms. We use these features of even and odd functions to obtain the half-range Fourier cosine and sine series, respectively.

Half range Fourier cosine series

Suppose that a half-range Fourier cosine series is desired for a function $f(x)$ which is defined on the interval $[0, L]$ as shown in the following figure.

Figure



Fourier series

Since a Fourier cosine series is obtained only for an even function, therefore we extend the function $f(x)$ over the other half of the interval $[-L, 0]$ so that it becomes an even function. With this extension, $f(x)$ is an even function defined on the interval $[-L, L]$. Then extend this new function as a periodic function of period $2L$ on the x -axis, as shown in the following figure.

Figure

We can now represent this extended function by a Fourier cosine series given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{L} x \quad (43)$$

$$\text{where } a_0 = \frac{2}{L} \int_0^L f(x) dx,$$



Fourier series

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi}{L} x \, dx, \quad n = 1, 2, 3, \dots$$

Since the Fourier series (43) is based on the function $f(x)$ defined on the interval $[0, L]$ which is half of the interval of periodicity $[-L, L]$, therefore this series is called the half-range Fourier cosine series.

Half range Fourier sine series

Suppose that a half-range Fourier sine series is desired for a function $f(x)$ which is defined on the interval $[0, L]$ as shown in the following figure.

Figure



Fourier series

Since a Fourier sine series is obtained only for an odd function, therefore we extend the function $f(x)$ over the other half of the interval $[-L, 0]$ so that it becomes an odd function. With this extension, $f(x)$ is an odd function defined on the interval $[-L, L]$. Then extend this new function as a periodic function of period $2L$ on the x -axis, as shown in the following figure.

Figure

We can now represent this extended function by a Fourier sine series given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{L} x \quad (44)$$

$$\text{where } b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x \, dx, \quad n = 1, 2, 3, \dots$$



Fourier series

Since the Fourier series (44) is based on the function $f(x)$ defined on the interval $[0, L]$ which is half of the interval of periodicity $[-L, L]$, therefore this series is called the half-range Fourier sine series.

Particular case of period 2π

When the function $f(x)$ is defined over the interval $[0, \pi]$, the half-range Fourier cosine series corresponding to $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$\text{where } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx, \quad n = 1, 2, 3, \dots$$



Fourier series

and the half-range Fourier sine series corresponding to $f(x)$ is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where $b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx, \quad n = 1, 2, 3, \dots$



Fourier series

Example:- If $f(x) = x$ ($0 < x < L$), then find

- i) the half-range Fourier cosine series
- ii) the half-range Fourier sine series

Solution:- i)

The graph of $f(x)$ over the given interval $[0, L]$, its even extension over the interval $[-L, 0]$, and its periodic extension of period $2L$ is shown in the following figure.



Fourier series

Exercise

1. Find the least period of the following functions:

- i) $\sin 2x$ ii) $\cos \frac{2\pi x}{k}$
iii) $\sin \pi x$ iv) $\cos \frac{2n\pi x}{k}$

2. Find the Fourier series of the following functions:

i) $f(x) = x^2 \quad (-\pi < x < \pi)$

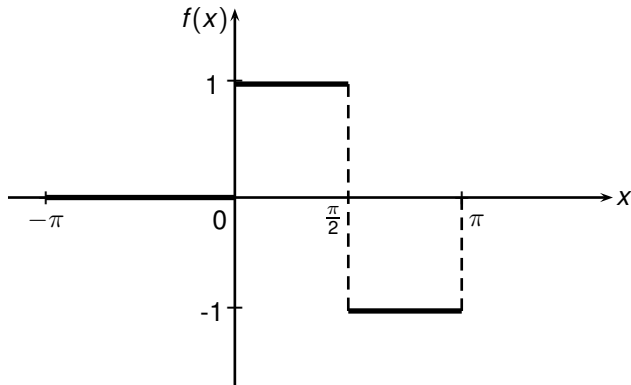
ii) $f(x) = x^3 \quad (-\pi < x < \pi)$

iii) $f(x) = e^x \quad (0 < x < 2\pi)$



Fourier series

3. Find the Fourier series of the following function.





Fourier series

4. Find the Fourier series of the following functions:

$$i) \quad f(x) = \begin{cases} x & \text{if } 0 < x < \pi \\ \pi - x & \text{if } \pi < x < 2\pi \end{cases}$$

$$ii) \quad f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } \pi < x < 2\pi \end{cases}$$

$$iii) \quad f(x) = \begin{cases} x & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \pi - x & \text{if } \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

5. Find the Fourier series of the following functions:

$$i) \quad f(x) = x^2, \quad 0 < x < 2, \quad P = 2L = 2$$

$$ii) \quad f(x) = 2|x|, \quad -2 < x < 2, \quad P = 2L = 4$$



Fourier series

$$iii) f(x) = \pi \sin \pi x, \quad 0 < x < 1, \quad P = 2L = 1$$

$$iv) f(x) = \begin{cases} \frac{1}{2} + x & \text{if } -\frac{1}{2} < x < 0 \\ \frac{1}{2} - x & \text{if } 0 < x < \frac{1}{2} \end{cases} \quad P = 2L = 1$$

6. Which of the following functions are odd, even, or neither odd nor even?

- | | |
|--------------------|-----------------------|
| i) $\sinh x$ | ii) $x \sin x$ |
| iii) $x^2 \cos nx$ | iv) $x x $ |
| v) $ x^3 $ | vi) $\sin x + \cos x$ |



Fourier series

7. Are the following functions $f(x)$ which are assumed to be periodic of period 2π , even, odd, or neither even nor odd?

$$i) \quad f(x) = \sin x \quad (-\pi < x < \pi)$$

$$ii) \quad f(x) = |x|, \quad (-\pi < x < \pi)$$

$$iii) \quad f(x) = \begin{cases} 0 & \text{if } -\pi < x < 0 \\ x & \text{if } 0 < x < \pi \end{cases}$$

8. Find the Fourier series of the following functions:

$$i) \quad f(x) = \begin{cases} -x^2 & \text{if } -\pi < x < 0 \\ x^2 & \text{if } 0 < x < \pi \end{cases}$$



Fourier series

$$ii) \quad f(x) = |\sin x| \quad (-\pi < x < \pi)$$

$$iii) \quad f(x) = x \sin x \quad (-\pi < x < \pi)$$

$$iv) \quad f(x) = \begin{cases} x^2 & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \frac{\pi^2}{4} & \text{if } \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

9. If $f(x) = x^2$ ($0 < x < L$), then find

- i) the half-range Fourier cosine series, and
- ii) the half-range Fourier sine series.

10. If $f(x) = \sin x$ ($0 < x < \pi$), then find

- i) the half-range Fourier cosine series, and
- ii) the half-range Fourier sine series.